A Proof of Detlefs' Sixth-Power Characterisation of the 11-Rough Numbers, OEIS A008364

Adrian Perez Fontelles
Independent Researcher
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Abstract

In December 2011, G. Detlefs contributed to OEIS A008364, the 11-rough numbers, the conjecture that these are exactly the n with $n^6 \equiv 1$ or $169 \pmod{210}$. A comment of A. del Arte from January 2014 asserts the statement is true and sketches a finite check, but no proof is recorded and the comment is still labelled a conjecture. We prove it, and identify the two residues as forced rather than found: the image of the sixth-power map on $(\mathbb{Z}/210\mathbb{Z})^\times$ is a subgroup of order $\prod_{p|210}(p-1)/\gcd(6, p-1) = 2$, so exactly two residues can occur, and the Chinese remainder theorem names them 1 and 169. The same computation explains why the companion conjecture on A008365 fails, needing five residues where only four are listed.

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1 The conjecture

OEIS A008364 is the sequence of 11-rough numbers, the positive integers not divisible by 2, 3, 5 or 7, that is those coprime to $210 = 2 \cdot 3 \cdot 5 \cdot 7$ [1]:

$$1,\ 11,\ 13,\ 17,\ 19,\ 23,\ 29,\ 31,\ 37,\ 41,\ 43,\ 47,\ 53,\ 59,\ 61, \ldots$$

On 30 December 2011, Gary Detlefs contributed:

> *Conjecture: these are numbers n such that* ($n^6 \bmod 210 = 1$) *or* ($n^6 \bmod 210 = 169$).

On 12 January 2014, Alonso del Arte added that the statement is true and "extremely easy to verify", proposing that one compute the sixth powers of the terms up to 209 and then check the complement. That is a valid finite verification — the condition depends only on $n \bmod 210$ — but no argument is recorded on the entry, the comment remains labelled a conjecture, and the prescription gives no reason why the answer set should have exactly two elements. Both comments were present on the live entry on 22 August 2026.

Theorem 1. *For every integer n,*

$$\gcd(n, 210) = 1 \iff n^6 \bmod 210 \in \{1,\ 169\},$$

and both residues occur, with least positive witnesses $n = 1$ and $n = 13$.

2 The general fact

Theorem 2. *Let $P = p_1 p_2 \cdots p_k$ be a product of distinct primes and let $e \geq 1$. Put*

$$H_e \;=\; \bigl\{\, m^e \bmod P \;:\; \gcd(m,P)=1 \,\bigr\}.$$

Then H_e is a subgroup of $(\mathbb{Z}/P\mathbb{Z})^\times$ of order

$$|H_e| \;=\; \prod_{i=1}^{k} \frac{p_i - 1}{\gcd(e,\, p_i - 1)},$$

and for every integer n, $\gcd(n,P)=1$ if and only if $n^e \bmod P \in H_e$.

Proof. By the Chinese remainder theorem $(\mathbb{Z}/P\mathbb{Z})^\times \;\cong\; \prod_i (\mathbb{Z}/p_i\mathbb{Z})^\times$, a direct product of cyclic groups of orders $p_i - 1$. In a cyclic group of order d the e-th power map is an endomorphism whose kernel has order $\gcd(e,d)$, so its image is the unique subgroup of order $d/\gcd(e,d)$; the image of a product of maps is the product of the images. If $\gcd(n,P)=1$ then $n^e \bmod P \in H_e$ by definition; conversely every element of H_e is a unit modulo P, so $n^e \bmod P \in H_e$ forces n^e, hence n, to be coprime to P. □

3 Proof of Theorem 1

Proof. Take $P = 210$ and $e = 6$ in Theorem 2. The prime factors are $2,3,5,7$ with $p-1$ equal to $1,2,4,6$, so

$$|H_6| \;=\; \frac{1}{\gcd(6,1)} \cdot \frac{2}{\gcd(6,2)} \cdot \frac{4}{\gcd(6,4)} \cdot \frac{6}{\gcd(6,6)} \;=\; 1 \cdot 1 \cdot 2 \cdot 1 \;=\; 2.$$

So exactly two residues can occur, and it remains to name them. For $p \in \{2,3,7\}$ we have $(p-1) \mid 6$, hence $n^6 \equiv 1 \pmod{p}$ for every n coprime to 210, and therefore

$$n^6 \equiv 1 \pmod{42}.$$

Modulo 5, $n^4 \equiv 1$ gives $n^6 \equiv n^2$, and the squares in $(\mathbb{Z}/5\mathbb{Z})^\times$ are $\{1,4\}$, both attained. Writing the residue as $x = 1 + 42t$ with $0 \leq t \leq 4$ and using $42 \equiv 2 \pmod{5}$, we get $x \equiv 1 + 2t \pmod{5}$; the requirement $x \equiv 1$ or $4 \pmod{5}$ gives $t = 0$ or $t = 4$, that is

$$x \in \{1,\; 169\}.$$

Hence $H_6 = \{1, 169\}$, and Theorem 2 supplies the equivalence. Both residues occur: $1^6 = 1$ and $13^6 = 4826809 = 22984 \cdot 210 + 169$. □

Remark 3. The proof shows the two residues are not a numerical accident. The prime 5 is the only one dividing 210 with $(p-1) \nmid 6$, and $\gcd(6,4) = 2$, so it alone contributes a factor of 2 to $|H_6|$. Every other prime is invisible at exponent 6.

Remark 4 (the companion conjecture). The same contributor stated, on the same day, the analogue for A008365, the 13-rough numbers: that they are the n with $n^{24} \bmod 2310$ in $\{1, 421, 631, 841\}$. There $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$ and

$$|H_{24}| = \frac{1}{1} \cdot \frac{2}{2} \cdot \frac{4}{4} \cdot \frac{6}{6} \cdot \frac{10}{2} = 5,$$

because 11 is the one prime with $(p-1) \nmid 24$. No four-element set can be correct, and indeed the missing residue is 1681, attained already at $n = 17$, the third term of that sequence. The contrast is instructive: at exponent 6 modulo 210 there are two residues, few enough to be found by hand; at exponent 24 modulo 2310 there are five, and one was dropped. The exponent that makes the answer a single residue is $\operatorname{lcm}(p-1) = 60$, not 24.

4 Computational verification

A short script (Python 3, standard library plus `sympy`) checks the following; all pass.

1. The image of the sixth-power map on $(\mathbb{Z}/210\mathbb{Z})^\times$, computed by exhaustion over all 48 units, is exactly $\{1, 169\}$.
2. The order formula of Theorem 2 agrees with brute-force enumeration for the primorials 2, 6, 30, 210, 2310 and the exponents $e \in \{2, 4, 6, 8, 10, 12, 24, 30, 60\}$.
3. For comparison, the image of the 24th power map on $(\mathbb{Z}/2310\mathbb{Z})^\times$ is $\{1, 421, 631, 841, 1681\}$, of size 5.

Every numeral in this note is an output of that script.

5 Concluding remarks

The result is elementary and was already believed true on the entry; what is added here is a proof rather than a verification recipe, and an explanation of the number 2. It is recorded because the comment has carried the label "Conjecture" since 2011, and because the same computation decides the companion statement on A008365 in the opposite direction. The natural destination is a comment on the entry rather than a journal.

We have found no earlier proof on the entry or in the literature, but an argument this short could well have been given before, and we would welcome notice of any earlier appearance.

References

[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`, 2026. Sequences A008364 (11-rough numbers) and A008365 (13-rough numbers); comments of G. Detlefs, 30 December 2011, and of A. del Arte, 12 January 2014.